```
Clear["Global`*"]
```

```
m1 = {{1, 0}, {-1/f1, 1}}
```

$$\left\{\{1, 0\}, \left\{-\frac{1}{f1}, 1\right\}\right\}$$

```
m2 = {{1, t}, {0, 1}}
```

$$\{\{1, t\}, \{0, 1\}\}$$

```
m3 = {{1, 0}, {-1/f2, 1}}
```

$$\left\{\{1, 0\}, \left\{-\frac{1}{f2}, 1\right\}\right\}$$

```
m4 = {{1, t}, {0, 1}}
```

$$\{\{1, t\}, \{0, 1\}\}$$

```
m5 = {{1, 0}, {-1/f1, 1}}
```

$$\left\{\{1, 0\}, \left\{-\frac{1}{f1}, 1\right\}\right\}$$

```
result = m5.m4.m3.m2.m1
```

$$\left\{\left\{1 - \frac{t}{f2} - \frac{t + t\left(1 - \frac{t}{f2}\right)}{f1}, t + t\left(1 - \frac{t}{f2}\right)\right\}, \left\{-\frac{1}{f1} - \frac{1 - \frac{t}{f1}}{f2} - \frac{1 - \frac{t}{f1} + t\left(-\frac{1}{f1} - \frac{1 - \frac{t}{f1}}{f2}\right)}{f1}, 1 - \frac{t}{f1} + t\left(-\frac{1}{f1} - \frac{1 - \frac{t}{f1}}{f2}\right)\right\}\right\}$$

```
(*in order to see any object clearly through our three-
 lens setup the system should be equivalent to the
  following matrix: {{1, 2*t}, {0, 1}} (free space matrix)*)
```

```
(*match result matrix to free space matrix*)
```

```
FullSimplify[result[[1, 2]] == 2 * t]
```

$$\frac{t}{f2} == 0$$

```
(*t << f2*)
```

```
FullSimplify[result[[1, 1]] == 1]
```

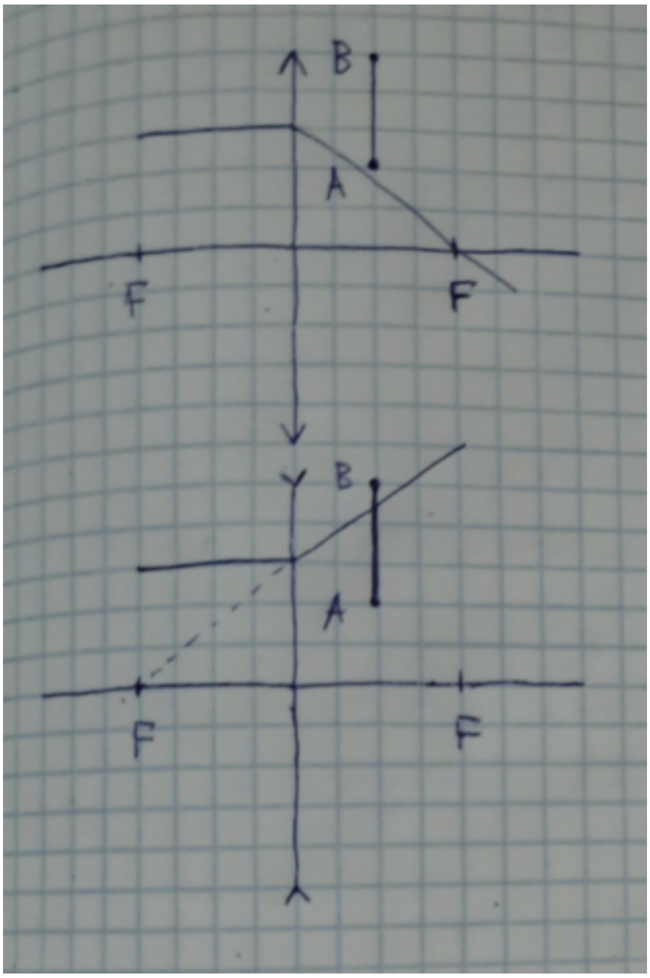
$$\frac{(f1 + 2\ f2 - t)\ t}{f1\ f2} == 0$$

```
(*throwaway t^2 since it's too small*)
```

```
Solve[(f1 + 2 * f2) / f1 / f2 == 0, f2]
```

$$\left\{\left\{f2 \to -\frac{f1}{2}\right\}\right\}$$

```
(*since the purpose of our three-lens setup is to hide any object
  (for example AB) between neighbour lenses side lenses will be converging*)
```




```
(*middle lens will be diverging since f2 seems to be negative*)

(*t << f2; f2 = f1/2 ===> t << f1/2 ===> t << f1*)

(*Q.E.D.*)
```